

Floods remain one of the most pervasive and costly natural hazards affecting communities worldwide, with their frequency and severity continuing to rise as global temperatures increase. According to the United Nations Office for Disaster Risk Reduction (UNDRR), flood events currently make up roughly 35 to 40 percent of all weather-related disaster occurrences globally, and the number of people exposed to flood risk has climbed steadily from 28.1 million in 1970 to 35.1 million in 2020, an increase of nearly 25 percent (UNDRR, 2025). This mounting exposure translates into substantial economic consequences, with global disaster-related costs now exceeding \$2.3 trillion annually once cascading and ecosystem impacts are accounted for (UNDRR, 2025).

The escalating scale of these impacts is closely tied to accelerating climate change: WMO's *State of Global Water Resources 2024* report confirms that 2024 was the hottest year in the 175-year observational record, with global surface temperatures reaching 1.55°C above pre-industrial levels, and notes that central, western, and eastern Africa, parts of Asia, and Central Europe experienced markedly wetter-than-normal conditions and flooding that year (WMO, 2025).

Within this global context, Bangladesh stands out as one of the most flood-vulnerable countries in the world. Situated at the outlet of the Ganges-Brahmaputra-Meghna delta, nearly 60% of its population faces elevated flood hazard, and roughly 45% is exposed to severe river flooding among the highest such figures globally (Rahman et al., 2025). This exposure is intensifying under climate change: rising sea levels and strengthening tropical cyclones are projected to sharply increase coastal storm-tide flooding along Bangladesh's coast in the coming decades (Qiu et al., 2025). Compounding these physical and climatic pressures, flood-prone communities often lack the institutional capacity and adaptive resources needed to cope, leaving vulnerability unevenly distributed across the population (Rahman et al., 2025). These intersecting geographic, climatic, and socioeconomic stressors make Bangladesh a critical setting for developing accurate, spatially explicit flood susceptibility assessments.

Flood susceptibility mapping provides a critical decision-support tool for policymakers, urban planners, and disaster management authorities, enabling them to prioritize infrastructure investment, guide land-use planning, and design targeted early-warning and emergency preparedness programs (Tesfaye et al., 2025). This predictive capability is especially valuable in transboundary or data-scarce basins, where susceptibility maps can assist practitioners in prioritizing interventions despite limited monitoring infrastructure (Ali et al., 2025). Accurately identifying flood-prone areas is therefore essential not only for immediate emergency response but also for long-term, sustainable resource allocation and risk reduction.

Land use and land cover (LULC) change, particularly urban expansion, and climate change together constitute the two major drivers of flood risk. As natural, permeable land cover is replaced with impervious surfaces such as pavement and rooftops, infiltration capacity declines sharply, increasing runoff volume and speed and elevating peak discharge, an effect that further

intensifies in rapidly urbanizing floodplains through population growth and encroachment into flood-prone areas (Ongaga et al., 2024; Hossain et al., 2026). Simultaneously, climate change intensifies extreme precipitation, as warming increases atmospheric moisture-holding capacity and enables convective rain systems to intensify beyond standard theoretical expectations, a trend already evident in the widespread, often record-breaking short-duration rainfall events observed globally in 2025 (Da Silva & Haerter, 2025; Green et al., 2026). Together, these human-driven and climatic factors underscore the need to explicitly incorporate both future land use and forward-looking climate projections, rather than static historical data alone, into flood susceptibility modeling.

Flood susceptibility assessment has traditionally relied on two broad approaches, each facing well-documented limitations. Qualitative, expert-driven methods such as the Analytical Hierarchy Process (AHP) and other multi-criteria decision-making (MCDM) techniques assign importance weights to flood-conditioning factors through structured expert judgment; however, this reliance on subjective pairwise comparisons introduces bias and inconsistency across studies, and such approaches struggle to represent the complex, nonlinear relationships that govern flood occurrence (Rizaliali et al., 2024). On the other hand, physically-based hydrodynamic models simulate flood processes using governing hydraulic equations, offering high physical realism, but this accuracy comes at the cost of intensive computational resources and dependence on detailed, high-resolution geospatial data that are frequently unavailable in data-scarce regions (Dahawi et al., 2026). Traditional approaches such as AHP-based MCDM and hydrodynamic simulation therefore face limitations including subjective weighting, inability to capture nonlinear factor interactions, and heavy computational and data demands, as discussed in the literature (Rizaliali et al., 2024; Dahawi et al., 2026).

In contrast, machine learning approaches directly address these limitations by learning flood-conditioning relationships from historical data rather than relying on subjective weighting or computationally intensive physical simulation. A range of ML algorithms have been applied to flood susceptibility mapping, including Support Vector Machine (SVM), Random Forest (RF), and eXtreme Gradient Boosting (XGBoost), which have been benchmarked together in multi-model comparative frameworks (Li et al., 2026), as well as deep learning architectures such as convolutional neural network-based autoencoders for regions with limited flood inventory data (Thappitla et al., 2025). Beyond model selection, hyperparameter optimization has emerged as a critical step for improving predictive performance: Li et al. (2026) demonstrated that a Particle Swarm Optimization-tuned XGBoost model (PSO-XGB) achieved the highest accuracy (0.933) and AUC (0.956) among all tested model-optimizer combinations, outperforming both non-optimized models and models tuned via Bayesian Optimization. This growing body of evidence indicates that machine learning, particularly when paired with metaheuristic hyperparameter tuning, offers superior flexibility, accuracy, and adaptability for flood susceptibility modeling compared to traditional approaches.

Despite this progress, important gaps remain in the Bangladesh context specifically. A recent comprehensive review of machine learning-based flood risk prediction in Bangladesh, synthesizing 42 influencing factors across the existing literature, explicitly identifies the integration of future climate projections into flood hazard mapping as an underexplored direction requiring further research (Islam et al., 2025). Some progress has been made in this direction: for instance, Rifath et al. (2024) combined RF, SVM, and XGBoost with CMIP6 climate projections (SSP1-2.6 and SSP5-8.5) to map present and future flash flood susceptibility in the hilly regions of southeastern Bangladesh. However, this and similar studies remain geographically confined to specific sub-regions and incorporate future climate scenarios alone, without concurrently projecting future land use and land cover (LULC) change, despite urbanization being a well-established driver of flood risk. Moreover, none of these studies apply systematic metaheuristic hyperparameter optimization, such as Particle Swarm Optimization, to their machine learning models. Although related work has separately examined ML-based flood susceptibility mapping in Bangladesh and combined ML with future climate projections at the sub-regional scale, few studies have integrated PSO-optimized machine learning models with both future LULC simulation and future climate projections at a national scale for Bangladesh, representing a gap that this study addresses.

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